Advanced Imaging of the Spine

Humana®
And Subsidiaries

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Medicare Advantage Medical Coverage Policy

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Disclaimer
The Medical Coverage Policies are reviewed by all Utilization Management Committees within Humana. Policies in this document may be modified by a member's coverage document. Clinical policy is not intended to preempt the judgment of the reviewing medical director or dictate to health care providers how to practice medicine. Health care providers are expected to exercise their medical judgment in rendering appropriate care. Identification of selected brand names of devices, tests and procedures in a medical coverage policy is for reference only and is not an endorsement of any one device, test, or procedure over another. Clinical technology is constantly evolving, and we reserve the right to review and update this policy periodically. References to CPT® codes or other sources are for definitional purposes only and do not imply any right to reimbursement or guarantee of claims payment. No part of this publication may be reproduced, stored in a retrieval system or transmitted, in any shape or form or by any means, electronic, mechanical, photocopying or otherwise, without permission from Humana and subsidiaries.

Related Medicare Advantage Medical/Pharmacy Coverage Policies

None

Related CMS Documents

Please refer to CMS Medicare Coverage Database for the most current applicable CMS National Coverage Determination (NCD)/Local Coverage Determination (LCD)/Local Coverage Article (LCA). Refer to CMS website for the most current applicable CMS Online Manual System (IOMs) and Transmittals.

Humana and subsidiaries apply a Medicare Administrative Contractor's (MAC's) LCD to all requests (both Part A and Part B services) within the MAC's applicable geographic jurisdiction.

Type	Title	Document ID Number	Jurisdiction Medicare Administrative	Applicable States/Territories

			Contractors (MACs)	
NCD	Computed Tomography	220.1		
NCD	Magnetic Resonance Imaging	220.2		
LCD LCA	Magnetic Resonance Angiography (MRA)	L33633 A56747	J6 - National Government Services, Inc. (Part A/B MAC) JK - National Government Services, Inc. (Part A/B MAC)	IL, MN, WI CT, MA, ME, NH, NY, RI, VT
LCD LCA	Lumbar MRI	L34220 A57206	JE - Noridian Healthcare Solutions, LLC JF - Noridian Healthcare Solutions, LLC	CA, HI, NV, American Samoa, Guam, Northern Mariana Islands AK, AZ, ID, MT, ND, OR, SD, UT, WA, WY
LCD LCA	Magnetic Resonance Angiography (MRA) Multiple Imaging in Oncology	L34865 A56805 L35391 A56848	JH - Novitas Solutions, Inc. (Part A/B MAC) JL - Novitas Solutions, Inc. (Part A/B MAC)	AR, CO, NM, OK, TX, LA, MS DE, D.C., MD, NJ, PA
LCD LCA	Magnetic Resonance Angiography (MRA)	L34424 A56775	JJ - Palmetto GBA (Part A/B MAC) JM - Palmetto GBA (Part A/B MAC)	AL, GA, TN NC, SC, VA, WV
LCD LCA	Magnetic Resonance Angiography (MRA)	L34372 A57779	JN - First Coast Service Options, Inc. (Part A/B MAC)	FL, PR, US VI

Description

Diagnostic imaging provides visualization of tissues and organs in the body to diagnose abnormalities and guide therapy. Techniques include, but may not be limited to:

- **Computed tomography (CT)** refers to a computerized x-ray imaging procedure in which a three-dimensional image (3D) of a body structure is revealed through a series of cross-sectional images or "slices." CT scans of the spine offer high quality representation of the bones or osseous details significant to degenerative disease, surgical planning and vertebral fractures.

- **Magnetic resonance angiography (MRA)** is a noninvasive diagnostic technique that produces computerized images of internal body tissues and is based on nuclear magnetic resonance of atoms within the body induced by the application of radio waves. Contrast agent (dye) may be injected into a vein to make the blood vessels and blood flow easier to see. MRA can be used to diagnose such conditions as spinal vascular malformations.

- **Magnetic resonance imaging (MRI)** is a noninvasive diagnostic technique that produces computerized images of internal body tissues and is based on nuclear magnetic resonance of atoms within the body induced by the application of radio waves. MRI of the spine is often used to detect such conditions as spinal cord abnormalities, infection or vascular disorders.

Coverage Determination

Humana and subsidiaries follow the Medicare requirements that only allow coverage and payment for services that are reasonable and necessary for the diagnosis and treatment of illness or injury or to improve the functioning of a malformed body member except as specifically allowed by Medicare.

Humana and subsidiaries apply any applicable National Coverage Determination (NCD) and any applicable Local Coverage Determinations (LCDs) to the services and jurisdiction at issue. See the Related CMS Documents Section above for any such NCDs or LCDs.

For jurisdictions without an LCD (J5/J8, J6/JK, J15, JE, JF, JJ/JM, JN), Humana and subsidiaries determine medical necessity for **general frequency guidance on the use of advanced imaging modalities (CT/MRI)** with the criteria contained in LCD – Multiple Imaging in Oncology (L35391).

In order to determine medical necessity consistently, Humana and subsidiaries may consider the criteria contained in the following:

<u>CT</u> (72125, 72126, 72127, 72128, 72129, 72130, 72131, 72132, 72133, 76380)

While NCD 220.1 Computed Tomography includes **computed tomography and multi-planar diagnostic imaging** as nationally covered tests, it does not provide criteria specific to when the use of **computed tomography** is considered medically reasonable and necessary. For all jurisdictions, Humana and subsidiaries may consider the following to interpret or supplement such criteria in order to determine medical necessity consistently:

CT of the spine (cervical, lumbar, thoracic) will be considered medically reasonable and necessary for the following:

- Back pain of the lumbar or thoracic spine with or without radiculopathy and any of the following conditions/symptoms (red flags[4,9,22]) identified in medical history and physical examination:
  - Acute onset of urinary retention or overflow incontinence
  - Bilateral or progressive weakness in the lower limbs
  - Constitutional symptoms (eg, fatigue, fever, unintentional weight loss)
  - History of cancer
  - Increased risk for fracture (eg, prolonged use of corticosteroid)
  - Increased risk for infection (eg, immunosuppression, intravenous [IV] drug use)
  - Loss of anal sphincter tone or fecal incontinence
  - Recent or current bacterial infection (eg, urinary infection)
  - Saddle anesthesia; **OR**

- Neck pain of the cervical spine with or without radiculopathy and any of the following conditions/symptoms (red flags[2]) identified in medical history and physical examination:
  - Coagulopathy
  - Constitutional symptoms (eg, fatigue, fever, unintentional weight loss)
  - Elevated inflammatory markers (eg, C- reactive protein [CRP], erythrocyte sedimentation rate [ESR], white blood cell [WBC] count)
  - Increased risk for fracture (eg, elderly, potentially osteoporotic)
  - Increased risk of infection (eg, immunosuppression, IV drug use)
  - Infection
  - Inflammatory arthritis
  - Malignancy
  - Vascular etiology suspected; **OR**

- Alignment abnormalities[10] for any of the following:
  - Kyphosis for surgical planning[27]; **OR**
  - Scoliosis for surgical planning or complications[28]; **OR**
  - Spondylolisthesis for definitive diagnosis of subtle fracture[29]; **OR**

- Cervical pain or cervical radiculopathy with any of the following[2]:
  - Prior cervical spine surgery, acute or increasing mechanical cervical pain or trauma[2]; **OR**
  - When MRI cannot be obtained for the following:
    - Suspected or known infection with acute or increasing cervical pain or radiculopathy[2]; **OR**
    - Diagnosis of malignancy with acute or increasing cervical pain or radiculopathy[2]; **OR**

- Degenerative changes (disc degeneration, spinal stenosis) when MRI cannot be obtained[10]; **OR**

- Imaging guidance for any of the following[10]:
  - Preoperative planning[10]; **OR**
  - Interventional procedures (eg, aspiration, biopsy, stereotactic) before, during and after[10]; **OR**
  - Postoperative evaluation (eg, fusion, implant position, instrumentation integrity, osseous detail)[10]; **OR**

- Infection, suspected (eg, epidural abscess or discitis osteomyelitis), initial imaging, when MRI cannot be obtained for any of the following[8]:
  - Back or neck pain, new or worsening with or without fever and one the following:
    - Any red flag condition[8]:
      - Diabetes
      - IV Drug use
      - Cancer
      - HIV
      - Dialysis; **OR**
    - Abnormal lab values (elevated CRP or ESR) [8]; **OR**
  - Recent intervention (surgery with or without hardware, pain injection, stimulator implantation)[8]; **OR**
  - Cauda equina syndrome or new neurologic deficit[8]; **OR**
  - Decubitus ulcer or wound overlying spine[8]; **OR**

- Inflammatory back pain, known or suspected axial spondyloarthritis with any of the following[3]:
  - Spine ankylosis, suspected fracture, initial imaging[3]; **OR**
  - Negative radiographs and negative MRI of the sacroiliac joints[3]; **OR**

- Low back pain with or without radiculopathy, initial imaging, when an MRI cannot be obtained for any of the following[4]:
  - Intervention or surgical candidate[4] with persistent or progressive symptoms despite four weeks[22] of conservative treatment with pharmacotherapy (eg, analgesics or muscle relaxants) or physical therapy[22,24,48]; **OR**
  - Prior history of lumbar surgery with new or progressing clinical findings/symptoms[4]; **OR**

- Myelopathy, acute onset, chronic or progressive when an MRI cannot be obtained, initial imaging[6]; **OR**

- Neoplastic conditions and complications for any of the following[10]:
  - Initial diagnosis workup, new or progressive symptoms, treatment response, follow up or surveillance[36,37,38]; **OR**
  - Central nervous system tumors when MRI cannot be obtained[10]; **OR**
  - Neuroblastoma- initial diagnosis, staging and treatment planning, post treatment assessment and surveillance[38]; **OR**
  - Suspected primary bone tumor found on radiograph[7]; **OR**

- Suspected primary bone tumor with clinical presentation (eg, worsening pain at night, muscle spasms, abnormal gait due to pain[30]) not found on radiograph[7]; **OR**

- Trauma with any of the following:
  - Blunt cervical or thoracolumbar, initial imaging[0]; **OR**
  - Follow up imaging for persistent cervical neck symptoms (eg, mild neurological findings, pain, point tenderness, reduced range of motion or stiffness) while individual remains in a collar brace with no unstable injury on initial imaging[0]; **OR**
  - Lower or thoracic back pain accompanying a minor fall or heavy lift in a potentially osteoporotic or elderly individual[4,9]; **OR**

- Vertebral compression fractures (VCF) with any the following[5]:
  - New back pain with history of previously treated or multiple VCFs[5]; **OR**
  - New symptomatic VCF identified on radiograph[5]

MRI (72141, 72142, 72146, 72147, 72148, 72149, 72156, 72157, 72158)
Please refer to CMS National Coverage Determination – Magnetic Resonance Imaging (220.2).

While the NCD – Magnetic Resonance Imaging (220.2) is well established for some indications, it is not fully established for all **evidence-based indications**. If a determination cannot be made based on the CMS guidance above (NCD 220.2) because such criteria is not fully established, Humana and subsidiaries may consider the following to interpret or supplement such criteria in order to determine medical necessity consistently:

For jurisdictions without an LCD (J5/J8, J6/JK, J15, JH/JL, JJ/JM, JN), Humana and subsidiaries determine medical necessity for **lumbar MRI (72148, 72149, 72158)** based on the criteria contained in LCD – Lumbar MRI (L34220).

In order to determine medical necessity consistently, Humana and subsidiaries may consider the criteria contained in the following:

Additionally, **the following applies to ALL jurisdictions**. While CMS provides some guidance for MRI, it does not provide criteria specific to when the use of an MRI for **pre/post procedure or treatment planning, treatment response, follow up or surveillance, repeat imaging** and all **evidence-based indications** are considered medically reasonable and necessary. Additionally, CMS does not provide **conservative treatments modalities used for low back pain.** Humana and subsidiaries may consider the following to interpret or supplement such criteria in order to determine medical necessity consistently:

MRI of the spine (cervical, lumbar, thoracic) will be considered medically reasonable and necessary for the following:

- Alignment disorders for bone, spinal cord or soft tissue abnormalities[27,28,29]; **OR**

- Cerebrospinal fluid leak[11]; **OR**

- Low back pain following 4 weeks[13,0] conservative treatment consisting of pharmacotherapy (eg, analgesics or muscle relaxants) or physical therapy[22]; **OR**

- Imaging guidance for any the following:

  - Initial procedure/treatment planning[11,36,37,38]; **OR**
  - New or progressive symptoms[11,36,37,38]; **OR**
  - Post treatment/procedure evaluation and surveillance[11,36,37,38]

MRA (72159, C8931, C8932, C8933)

While CMS provides some guidance for **MRA (NCD 220.2**, **L33633, L34865, L34424, L34372**), it does not provide criteria specific to when the use of an **MRA for the spinal canal** is considered medically reasonable and necessary. To support consistent application of medical necessity criteria across all jurisdictions, Humana and subsidiaries may consider the following to interpret or supplement existing CMS guidance:

MRA of the spinal canal will be considered medically reasonable and necessary when any of the following criteria are met:

- Suspicion of spinal vascular malformation (eg, arteriovenous malformation [AVM], spinal arteriovenous fistula [AVF])[12];**OR**

- Myelopathy is known and compromise of blood flow or drainage to the spinal cord is suspected[31]; **OR**

- Preoperative evaluation for procedure with possible impact to spinal vasculature for **ANY** of the following:

  - Aortic aneurysm repair[12]
  - Localization of the spinal arteries for complex spinal procedure[12]
  - Characterization of suspected vascular lesions, including tumors or masses of the spinal canal and its contents[12]; **OR**

- Follow-up study[12] for **ANY** of the following:

  - Known AVM[23];
  - Known AVF[32];
  - Post procedure, surgery or treatment progress (eg, aortic aneurysm repair, localization of the spinal arteries for complex spinal procedure) [32]; **OR34**

- Vertebral artery injury is suspected or known and vascular compromise to the spinal canal and its contents is suspected[34]

The use of the criteria above provides clinical benefits highly likely to outweigh any clinical harms (eg, adverse effects including, but not limited to, ionizing radiation or risk in the MR environment exposure[10,11]). Services that do not meet the criteria above are not medically reasonable and necessary and may result in unnecessary exposure to potential complications. Medically unnecessary services carry risks of adverse outcomes and may interfere with the pursuit of other treatments which have demonstrated efficacy.

Coverage Limitations

US Government Publishing Office. Electronic code of federal regulations: part 411 – 42 CFR § 411.15 - Particular services excluded from coverage

Please refer to CMS guidance regarding advanced imaging of the spine.

Coding Information

Any codes listed on this policy are for informational purposes only. Do not rely on the accuracy and inclusion of specific codes. Inclusion of a code does not guarantee coverage and/or reimbursement for a service or procedure.

CPT® Code(s)	Description	Comments
72125	Computed tomography, cervical spine; without contrast material	
72126	Computed tomography, cervical spine; with contrast material	
72127	Computed tomography, cervical spine; without contrast material, followed by contrast material(s) and further sections	
72128	Computed tomography, thoracic spine; without contrast material	
72129	Computed tomography, thoracic spine; with contrast material	
72130	Computed tomography, thoracic spine; without contrast material, followed by contrast material(s) and further sections	
72131	Computed tomography, lumbar spine; without contrast material	
72132	Computed tomography, lumbar spine; with contrast material	
72133	Computed tomography, lumbar spine; without contrast material, followed by contrast material(s) and further sections	

72141	Magnetic resonance (eg, proton) imaging, spinal canal and contents, cervical; without contrast material	
72142	Magnetic resonance (eg, proton) imaging, spinal canal and contents, cervical; with contrast material(s)	
72146	Magnetic resonance (eg, proton) imaging, spinal canal and contents, thoracic; without contrast material	
72147	Magnetic resonance (eg, proton) imaging, spinal canal and contents, thoracic; with contrast material(s)	
72148	Magnetic resonance (eg, proton) imaging, spinal canal and contents, lumbar; without contrast material	
72149	Magnetic resonance (eg, proton) imaging, spinal canal and contents, lumbar; with contrast material(s)	
72156	Magnetic resonance (eg, proton) imaging, spinal canal and contents, without contrast material, followed by contrast material(s) and further sequences; cervical	
72157	Magnetic resonance (eg, proton) imaging, spinal canal and contents, without contrast material, followed by contrast material(s) and further sequences; thoracic	
72158	Magnetic resonance (eg, proton) imaging, spinal canal and contents, without contrast material, followed by contrast material(s) and further sequences; lumbar	
72159	Magnetic resonance angiography, spinal canal and contents, with or without contrast material(s)	
76380	Computed tomography, limited or localized follow-up study	
CPT® Category III Code(s)	**Description**	**Comments**
No code(s) identified		
HCPCS Code(s)	**Description**	**Comments**
C8931	Magnetic resonance angiography with contrast, spinal canal and contents	
C8932	Magnetic resonance angiography without contrast, spinal canal and contents	
C8933	Magnetic resonance angiography without contrast followed by with contrast, spinal canal and contents	

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Change Summary

05/01/2025 New Policy
12/01/2025 Update, Coverage Change. Title Change. Updated Coding Information
01/02/2026 Update, No Coverage Change